Fiscal Year:	FY 2025	**Task Last Updated:**	FY 06/30/2025
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Project Title:	AS-PHYS-Bloch Oscillations in Microgravity LOI		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	Other names: CA Sackett; Cass Sackett.		
Project Type:	Flight	**Solicitation / Funding Source:**	2022 Physical Sciences NNH22ZDA001N-FP E.6 Fundamental Physics
Start Date:	06/05/2024	**End Date:**	09/30/2029
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	3
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA HQ
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Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 09/30/2029 per J. Callas/JPL (Ed., 6/24/25). NOTE: End date changed to 09/30/2028 per D. Griffin/NASA-HQ. The original period of performance was 06/05/24 - 06/04/28 (Ed., 1/21/25).		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	80NSSC24K1237		
Performance Goal No.:			
Performance Goal Text:			

Task Description:

The upgrade of the NASA Cold Atom Laboratory (CAL) provides the opportunity for new flight experiments. We propose here to explore the phenomenon of Bragg oscillation in a microgravity environment, with applications to precise and robust measurements of small accelerations at the level of 1 cm/s^2. A parallel set of ground experiments will help guide the flight effort.
Bloch oscillations occur when atoms are held in an optical lattice potential, in the presence of an accelerating force pointing along the lattice direction. The force causes the atoms to tunnel through the lattice barriers, so the atoms accelerate much as they would in free space. However, when the velocity of the atoms reaches the Bragg velocity of the lattice, the atoms are coherently reflected and their momentum is reversed. The accelerating force continues to act and the momentum increases again, with the cycle repeating as long as the atoms can be retained. Bloch oscillations are an interesting physical phenomenon with many applications. For example, by measuring the period of the oscillation, the strength of the acceleration can be determined with high precision [1]. [Ed. Note: See References.]

In terrestrial experiments, the acceleration is typically due to gravity and the amplitude of the oscillation is on the order of micrometers, since the atoms do not fall very far before reaching the Bragg velocity of order 1 cm/s. In microgravity, the oscillation amplitude is much larger, reaching the mm scale at milli-g accelerations. A few terrestrial experiments have investigated Bloch oscillations at low accelerations using horizontal optical lattices. However, these efforts are hampered by the need to support the atoms against gravity. This requires a combination of a relatively tight optical trap and magnetic levitation. The tight trap leads to high atomic density and thus complicating effects from interactions. Magnetic levitation cannot easily be made uniform, leading to spatial variations in the accelerating force. For instance, the experiments by Geiger et al. [2] observed oscillation amplitudes of about 150 um, but required a Feshbach resonance to nullify interactions and were limited by inhomogeneity. The sensitivity to accelerations increases with the amplitude of oscillation, so larger amplitudes are desirable.

The microgravity environment on CAL offers a clear opportunity for improvement. The existing Bragg laser can be used to provide the optical lattice, where a depth of about one recoil energy is needed. A tunable acceleration can be obtained using the magnetic gradient control provided in the new upgrade. A gradient of 15 mG/cm gives a suitable acceleration of 1 cm/s^2. The period of the oscillation would be about 0.5 s, which is within the vacuum limits of the system.

Although Bloch oscillations can be described using classical language, they are in fact a quantum phenomenon; classical particles would not "Bragg reflect" from a lattice. The amplitude of the oscillation is set by the spatial extent of the Wannier-Stark wavefunctions of the lattice, and the atomic waves are coherent across this length scale. Large Bloch oscillations are thus comparable to atom interferometry, but Bloch oscillations can be more robust to technical imperfections. For instance, they do not require a well-defined initial atom velocity, and they can tolerate a range of lattice depths. Our group is well-situated to carry out ground studies to support the flight operations. We have an existing apparatus that provides weakly-confined atoms in a highly uniform horizontal magnetic guide. We propose to use this system to test the approach and technical requirements, so that the flight experiments can be more quickly achieved and optimized.

References

[1] Nalcz I, Masi L, Ferioli G, Petrucciani T, Fattori M, Chwedenczuk J. Sensitivity bounds of a spatial Bloch-oscillation atom interferometer. Physical Review A. 2020 Sep;102(3):033318. https://

[2] Geiger ZA, Fujiwara KM, Singh K, Senaratne R, Rajagopal SV, Lipatov M, Shimasaki T, Driben R, Konotop VV, Meier T, Weld DM. Observation and uses of position-space Bloch oscillations in an ultracold gas. Physical review letters. 2018 May 25;120(21):213201. https://

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

The primary impact of the research will be on the development of tools for use in further ultracold atom research in microgravity. In particular, the work will demonstrate improved atom cooling techniques, and it will develop Bloch oscillations as a tool for atom interferometry. These techniques offer unique opportunities in microgravity, but will work differently than in a terrestrial environment. Long-term applications for these techniques include inertial sensing, gravity mapping, and tests of Einstein’s equivalence principle.

Direct progress on the proposed work has been impossible due to technical problems with the NASA Cold Atom Laboratory (CAL) apparatus. The proposed work requires capabilities that were planned to be implemented on the Servicing Mission 3B (SM3B) science module, which was launched in late 2023. Unfortunately, the apparatus suffered a vacuum leak prior to installation, rendering it unusable. Jet Propulsion Laboratory (JPL) and NASA were able to quickly launch a replacement in Spring 2024, using the SM1 science module that had previously been serving as a ground test bed. However, SM1 lacks several capabilities required for the proposed work, including “meso chips” for improved atom number and magnetic field control, and the Bragg laser beam required for atom interferometry and Bloch oscillation studies.
Over the performance period, the Principal Investigator (PI) has been working with JPL to support the preparation of another upgrade unit, SM3X, which will implement the meso chip and atom interferometry features originally planned for SM3B. In addition, JPL has identified a source of stray magnetic fields near the atoms, the feedthrough pins used to make electrical connections to the atom chip. These have been replaced with non-magnetic pins in SM3X. Stray magnetic fields have placed significant constraints on adiabatic cooling in previous versions of CAL, so we are excited that SM3X may permit even better cooling performance than initially planned. At this time, the SM3X module is scheduled to launch in April 2026.

With the advanced capabilities unavailable during the current performance period, we have focused efforts on alternative goals. An initial effort was to demonstrate simultaneous adiabatic cooling of rubidium and potassium atoms. This will be an important technique for future equivalence principle tests, and has been a major component in our previous projects. It has proven challenging because it has been difficult to reliably obtain simultaneous ultracold samples of both atomic species. Unfortunately, this challenge continued with SM1, and progress was limited until new defects in the apparatus developed in January 2025 which now largely prevent the production of any ultracold gases.

Despite this run of bad luck, we are continuing to use the apparatus for interesting work. SM1 is still capable of

Task Progress:	producing cold gases, with temperatures in the microKelvin range, and of confining the gas in a magnetic trap. Recent experiments have shown that a large number of atoms can be trapped this way, and that the trapped-atom lifetime is long enough to implement evaporative cooling. In previous work on CAL, all evaporative cooling was carried out with atoms trapped on the atom chip, which is now not functioning. Here instead we consider cooling in a “quadrupole” trap produced by CAL’s electromagnetic coils. Although the quadrupole trap is unsuitable for producing Bose-Einstein condensation, studies of cooling in it are interesting for several reasons. First, an initial cooling stage in the quadrupole trap could improve the efficiency of loading atoms onto the chip trap, enabling improved performance in SM3X and future experiments. Second, the quadrupole trap provides a simple configuration to compare the performance of evaporative cooling in microgravity and on Earth, by comparing the results of identical experiments on CAL and in the ground test bed. Although this comparison is of great technical interest, it has not yet been carried out using CAL. Finally, an atom-loss mechanism specific to quadrupole traps is the Majorana transition, in which atoms encountering a zero of the magnetic field undergo a spin flip and are ejected from the trap. On Earth, gravity distorts the distribution of atoms in the trap and complicates comparison to theoretical models. Microgravity removes this complication, making it an interesting system to investigate. Furthermore, while atoms undergoing Majorana loss quickly fall out of the trap in gravity, on CAL it may be possible to observe the untrapped atom cloud, and thereby investigate the spin distribution produced by the Majorana process. In Spring 2025, we carried out preparatory work for these experiments, including implementation of quadrupole trapping routines, characterization of atom number and temperature, and a preliminary measurement of the trapped atom lifetime. Currently, the apparatus is being assessed to allow longer hold times in the quadrupole trap, which will facilitate the planned work until SM3X is available. In addition to this work center on CAL, we have also carried out ground-based work to advance atom interferometry techniques. During the current period, efforts have been focused on rebuilding our interferometry apparatus to use atom chips, similar to the technique used on CAL. The new apparatus will support high-performance interferometers in chip-based traps, and allow rapid characterization and comparison of different atom chip geometries. This work will support future microgravity missions.
Bibliography Type:	Description: (Last Updated: 04/22/2026)
Abstracts for Journals and Proceedings	Sackett CA. "Atom interferometry experiments on the International Space Station." 2025 IEEE International Frequency Control Symposium and European Frequency and Time Forum, Queretaro, Mexico, May 12-16, 2025. Abstracts. 2025 IEEE International Frequency Control Symposium and European Frequency and Time Forum, Queretaro, Mexico, May 12-16, 2025. , May-2025
Abstracts for Journals and Proceedings	Sackett CA. "Quantum sensing using trapped atom interferometry." Quantum Days 2025, Toronto, Canada, February 19-21, 2025. Abstracts. Quantum Days 2025, Toronto, Canada, February 19-21, 2025. , Feb-2025